

Question	Marking Guidance	Mark	Comments
06.1	1. Pathogens OR Cells from an organism of a different species; 2. Cells from other organisms of the same species; 3. Abnormal body cells; 4. Antigen-presenting cells;	2 max (2 x AO1)	Reject toxins 1. Accept named examples of pathogens 1. Accept bacteria/fungi 1. Ignore viruses 2. Accept named appropriate cells from other organisms of the same species 2. and 3. Ignore B cells/T cells 3. Accept cancer cells 3. Accept cell infected with virus

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06.2	As a control (experiment), to show that it is OXA affecting the (immune) response OR As a control (experiment), to show that (olive) oil is not affecting the (immune) response OR To use as a control/standard/reference/starting point, to compare with (after) OXA (exposure);	1 (AO3)	Reject 'control/controlled variable'

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06.3	1. Labelled axes correct way round, linear scale and units; 2. Mean points plotted correctly; 3. SD bars correctly plotted above <u>and</u> below the peak of each bar;	3 (3 x AO2)	1. Reject if line graph drawn 1. Reject if Y-axis does not cover at least half of the grid 1. Reject if bars not of equal width 1. Accept a dual bar chart drawn 1. Reject if bars are touching (except dual bars) 1. Accept interruption drawn on the y axis 2. and 3. Allow all plots to the nearest half cm

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06.4	Cellular response 1. Female to female no significant difference in cellular response as SD overlap; 2. Male to male no significant difference in cellular response as SD overlap; 3. Significant increase in cellular response in autoimmune male compared with autoimmune female as SD do not overlap Humoral response 4. Male to male no significant difference in humoral response as SD overlap; 5. Female to female significant increase in humoral response as SD do not overlap; 6. Significant increase in humoral response in autoimmune female compared with autoimmune male as SD do not overlap	3 max (3 x AO3)	Max 2 for answers only relating to the cellular response or humoral response Accept '(ear) thickness' for cellular response, and 'concentration of anti-OXA/antibody' for humoral response If no other marks awarded, accept 1 principle mark for the idea that if SD overlap there is no significant difference or the converse 1. 2. and 4. Accept difference (likely) due to chance for no significant effect 3. 5. and 6. Accept increase not (likely) due to chance for significant increase Allow 'error bars' for 'SD'

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06.5	Supporting (Oestrogen) increases the humoral response that produces antibody; - More antibodies could increase progression of SLE; (Oestrogen) decreases the cellular response that produces T_C cells; - Fewer T_C cells could decrease/slow progression of RA; - Mice and humans are both mammals, so likely to have similar effects in both; Against - Increase in response might mean quicker production of antibody (not more) OR Decrease in response might mean slower production of T_C cells (not fewer); - Decrease in cellular response could (also) mean fewer antigen-presenting cells (and not just T_C cells); (Investigation) done in mice/not humans; Table 3 does not state which type of autoimmune disease the mice had OR Mice might not suffer from SLE/RA;	4 max (4 x AO3)	Max 3 for reasons supporting or against

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06.6	No – no mark 1. Mice with autoimmune disease will be unlikely to reproduce/survive OR Mice with autoimmune disease will be selected against; 2. Will not pass on allele (for autoimmune disease) OR Allele frequency (for autoimmune disease) will reduce/change; Yes – no mark 3. As long as the autoimmune disease did not affect the mice's ability to reproduce/survive; 4. The allele frequency will remain constant/not change;	2 max (2 x AO2)	Award as mark points 1 and 2, OR 3 and 4